

Department of
COMPUTER SCIENCE & ENGINEERING



ASSIGNMENT

Course Name: Microprocessors & Microcontrollers Sessional
Course Code: CSE 3212

Submitted By:

Md. Amimul Ihsan

Lecturer,

Department of EEE,
JSTU

Submitted By:

Name : **Monir Ahmed**

ID : 22111120

Session : 2021-2022

Department of CSE, JSTU

Submission Date: 19.04.2026

Chapter 4 Problems

Problem No. 4.8

Problem Statement:

Write a program to (a) display a "?", (b) read two decimal digits whose sum is less than 10,

(c) display them and their sum on the next line, with an appropriate message.

Sample execution:

?27

THE SUM OF 2 AND 7 IS 9

Code:

```
.model small
.stack 100h

.data
msg1 db '?$'
msg2 db 13,10,'THE SUM OF $'
msg3 db ' AND $'
msg4 db ' IS: $'
msg5 db 13,10,'INVALID (SUM >= 10)$'

.code
main:
    mov ax, @data
    mov ds, ax

    ; print ?
    mov dl, '?'
    mov ah, 02h
    int 21h

    ; input 1st digit
    mov ah, 01h
    int 21h
    sub al, '0'
    mov bl, al

    ; input 2nd digit
    mov ah, 01h
    int 21h
    sub al, '0'
    mov bh, al

    ; sum
```

```
mov al, bl
add al, bh
cmp al, 10
jge invalid
```

```
; print: THE SUM OF
lea dx, msg2
mov ah, 09h
int 21h
```

```
; print first number
mov dl, bl
add dl, '0'
mov ah, 02h
int 21h
```

```
; print " AND "
lea dx, msg3
mov ah, 09h
int 21h
```

```
; print second number
mov dl, bh
add dl, '0'
mov ah, 02h
int 21h
```

```
; print " IS "
lea dx, msg4
mov ah, 09h
int 21h
```

```
; print sum
add bl, bh
mov dl, bl
add dl, '0'
mov ah, 02h
int 21h
```

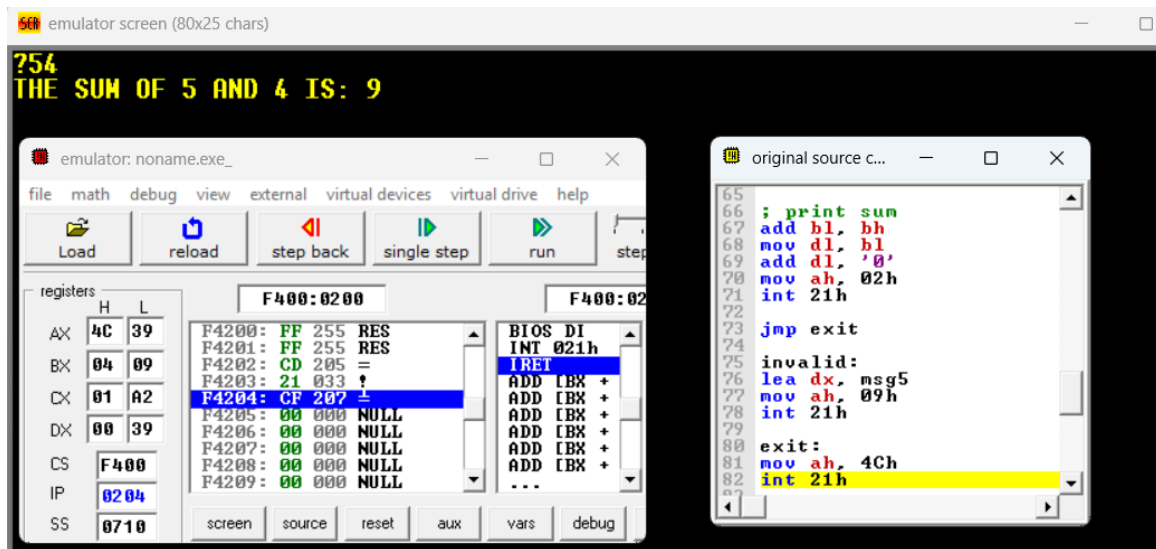
```
jmp exit
```

```
invalid:
    lea dx, msg5
    mov ah, 09h
    int 21h
```

```
exit:
    mov ah, 4Ch
    int 21h
```

```
end main
```

Output:



Problem No. 4.9

Problem Statement:

Write a program to (a) prompt the user, (b) read first, middle, and last initials of a person's name, and (c) display them down the left margin.

Sample execution:

ENTER THREE INITIALS: JFK

J

F

K

Code:

```
.model small
.stack 100h

.code
main:
    mov ax, @data
    mov ds, ax

    ; prompt
    mov dx, offset msg
    mov ah, 09h
    int 21h

    ; read 1st initial
    mov ah, 01h
    int 21h
```

```
mov bl, al
```

```
; read 2nd initial  
mov ah, 01h  
int 21h  
mov bh, al
```

```
; read 3rd initial  
mov ah, 01h  
int 21h  
mov cl, al
```

```
; new line  
mov dl, 13  
mov ah, 02h  
int 21h  
mov dl, 10  
int 21h
```

```
; print 1st  
mov dl, bl  
mov ah, 02h  
int 21h
```

```
mov dl, 13  
int 21h  
mov dl, 10  
int 21h
```

```
; print 2nd  
mov dl, bh  
mov ah, 02h  
int 21h
```

```
mov dl, 13  
int 21h  
mov dl, 10  
int 21h
```

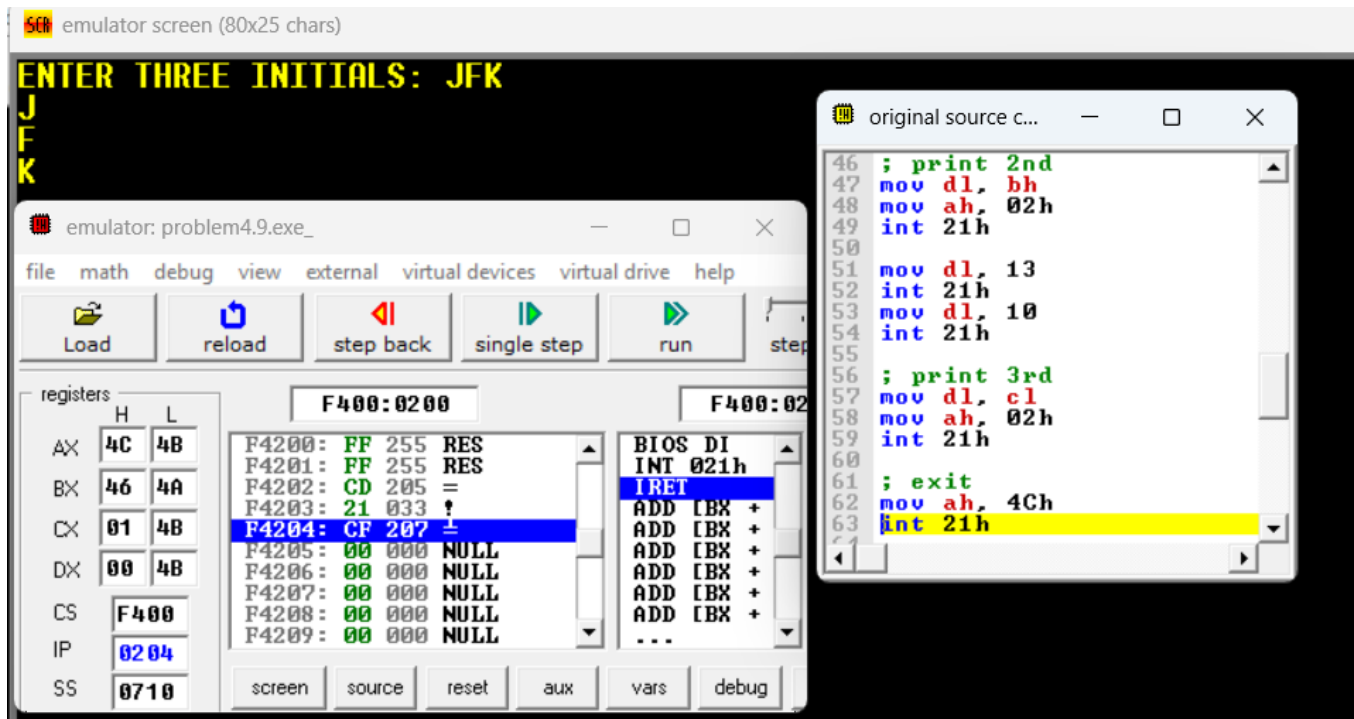
```
; print 3rd  
mov dl, cl  
mov ah, 02h  
int 21h
```

```
; exit  
mov ah, 4Ch  
int 21h
```

```
.data  
msg db 'ENTER THREE INITIALS: $'
```

```
end main
```

Output:



Problem No. 4.10

Problem Statement:

Write a program to read one of the hex digits A-F, and display it on the next line in decimal.

Sample execution:

ENTER A HEX DIGIT: C

IN DECIMAL IT IS 12

Code:

```
.model small
.stack 100h

.data
msg1 db 'ENTER A HEX DIGIT: $'
msg2 db 13,10,'IN DECIMAL IT IS $'

.code
main:
    mov ax, @data
    mov ds, ax

    ; prompt
    lea dx, msg1
    mov ah, 09h
    int 21h
```

```

; input
mov ah, 01h
int 21h

; convert A-F ? 10-15
sub al, 55      ; 'A'=65 ? 10

mov bl, al

; print message
lea dx, msg2
mov ah, 09h
int 21h

; print decimal (10-15)
mov al, bl
mov ah, 0
mov cl, 10
div cl          ; AL=1, AH=remainder

add al, '0'
mov dl, al
mov ah, 02h
int 21h

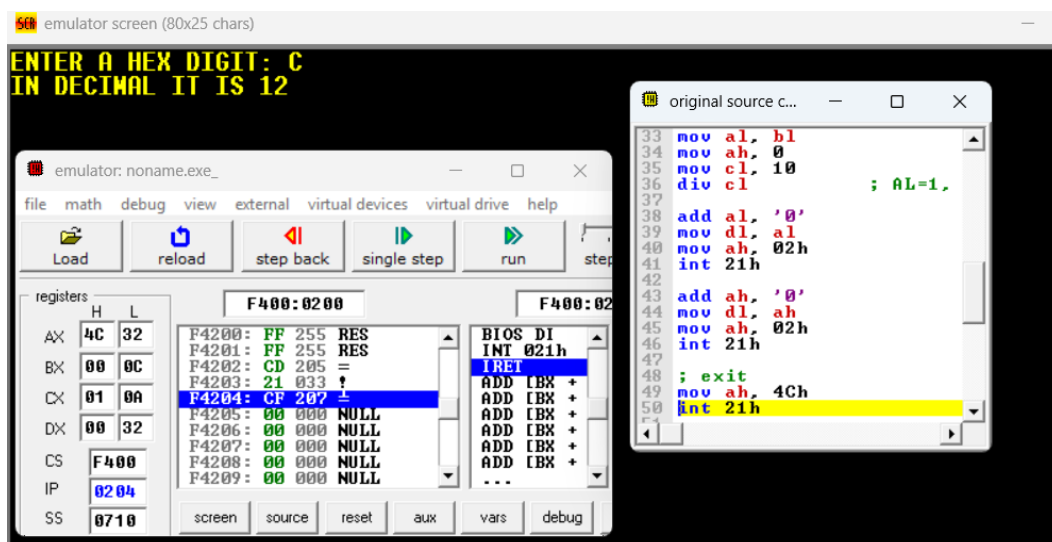
add ah, '0'
mov dl, ah
mov ah, 02h
int 21h

; exit
mov ah, 4Ch
int 21h

end main

```

Output:



Problem No. 4.11

Problem Statement:

Write a program to display a 10 x 10 solid box of asterisks.

Hint: declare a string in the data segment that specifies the box, and display it with INT 21h, function 9h.

Code:

```
.model small
.stack 100h

.data
box db '*****',13,10
    db '*****',13,10
    db '*****',13,10
    db '*****',13,10
    db '*****',13,10
    db '*****',13,10
    db '*****',13,10
    db '*****',13,10
    db '*****',13,10
    db '*****','$'

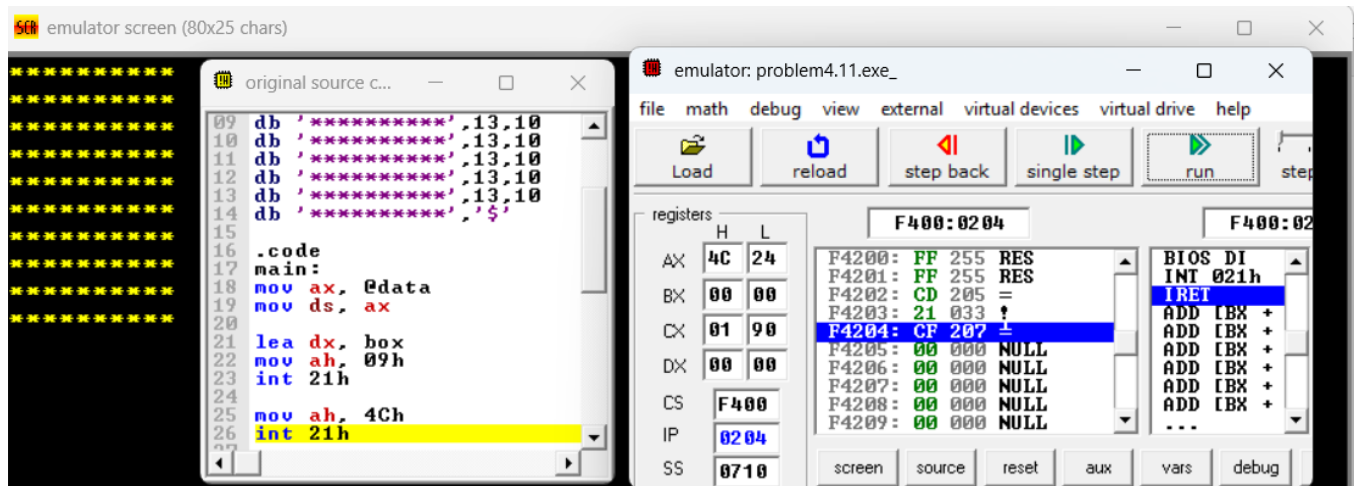
.code
main:
    mov ax, @data
    mov ds, ax

    lea dx, box
    mov ah, 09h
    int 21h

    mov ah, 4Ch
    int 21h

end main
```

Output:



Problem No. 4.12

Problem Statement:

Write a program to (a) display "?", (b) read three initials, (c) display them in the middle of an 11 x 11 box of asterisks, and (d) beep the computer.

Code:

```
.model small
.stack 100h

.data
row db '*****',13,10,'$'
c1 db ?
c2 db ?
c3 db ?

.code
main:
    mov ax, @data
    mov ds, ax

    ; display '?'
    mov dl, '?'
    mov ah, 02h
    int 21h

    ; read initials
    mov ah, 01h
    int 21h
    mov c1, al

    mov ah, 01h
    int 21h
    mov c2, al

    mov ah, 01h
    int 21h
    mov c3, al

    ; newline
    mov dl, 13
    mov ah, 02h
    int 21h
    mov dl, 10
    int 21h

    ; top 5 rows
    mov cx, 5
top:
    lea dx, row
    mov ah, 09h
    int 21h
```

```

loop top

; middle row ****ABC****
mov dl, '*'
mov ah, 02h
mov cx, 4
s1:
    int 21h
    loop s1

; print initials
mov dl, c1
int 21h
mov dl, c2
int 21h
mov dl, c3
int 21h

mov dl, '*'
mov cx, 4
s2:
    int 21h
    loop s2

; newline
mov dl, 13
int 21h
mov dl, 10
int 21h

; bottom 5 rows
mov cx, 5
bot:
    lea dx, row
    mov ah, 09h
    int 21h
    loop bot

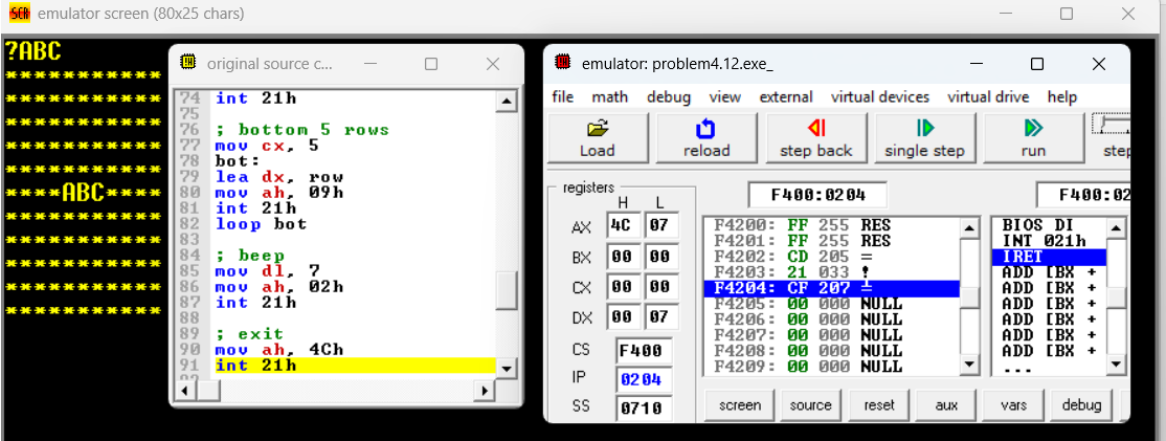
; beep
mov dl, 7
mov ah, 02h
int 21h

; exit
mov ah, 4Ch
int 21h

end main

```

Output:



Chapter 6 Problems

Problem No. 6.8

Problem Statement:

Write a program to display a "?", read two capital letters, and display them on the next line in alphabetical order.

Code:

```
.model small
.stack 100h
.data
msg db '?','$'
newline db 0Dh,0Ah,'$'

.code
main:
    mov ax, @data
    mov ds, ax

    ; display '?'
    mov ah, 09h
    lea dx, msg
    int 21h

    ; read first character
    mov ah, 01h
    int 21h
    mov bl, al

    ; read second character
    mov ah, 01h
    int 21h
    mov bh, al

    ; new line
    mov ah, 09h
    lea dx, newline
    int 21h

    ; compare and print in order
    cmp bl, bh
    jbe print_order

    ; swap
    mov al, bl
    mov bl, bh
    mov bh, al
```

```

print_order:
    ; print first
    mov dl, bl
    mov ah, 02h
    int 21h

    ; print second
    mov dl, bh
    mov ah, 02h
    int 21h

    mov ah, 4Ch
    int 21h
end main

```

Output:

```

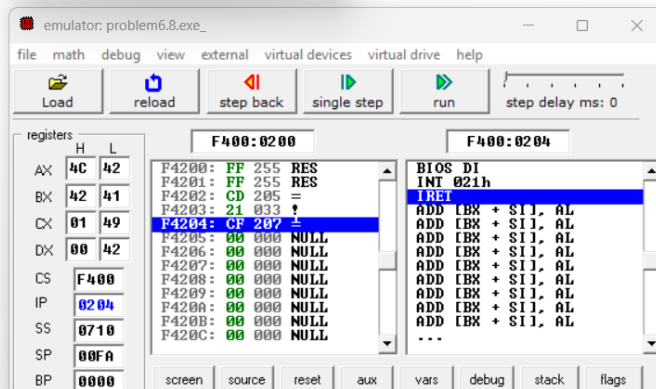
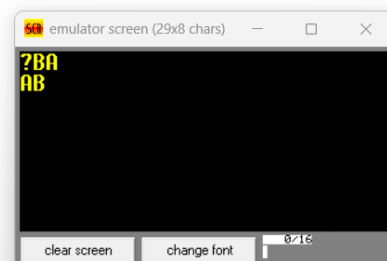
01 .model small
02 .stack 100h
03 .data
04 msg db '?','$',
05 newline db 0Dh,0Ah,'$'
06
07 .code
08 main:
09     mov ax, @data
10     mov ds, ax
11
12     ; display '?'
13     mov ah, 09h
14     lea dx, msg
15     int 21h
16
17     ; read first character
18     mov ah, 01h
19     int 21h
20     mov bl, al
21
22     ; read second character
23     mov ah, 01h
24     int 21h
25     mov bh, al
26
27     ; new line
28     mov ah, 09h
29     lea dx, newline
30     int 21h
31
32     ; compare and print in order
33     cmp bl, bh
34     jbe print_order
35
36     ; swap
37     mov al, bl
38     mov bl, bh
39     mov bh, al
40
41 print_order:
42     ; print first
43     mov dl, bl
44     mov ah, 02h
45     int 21h
46
47     ; print second
48     mov dl, bh
49     mov ah, 02h
50     int 21h
51
52     mov ah, 4Ch
53     int 21h
54 end main

```

```

36 ; swap
37 mov al, bl
38 mov bl, bh
39 mov bh, al
40
41 print_order:
42 ; print first
43 mov dl, bl
44 mov ah, 02h
45 int 21h
46
47 ; print second
48 mov dl, bh
49 mov ah, 02h
50 int 21h
51
52 mov ah, 4Ch
53 int 21h

```



Problem No. 6.9

Problem Statement:

Write a program to display the extended ASCII characters (ASCII codes 80h to FFh). Display 10 characters per line, separated by blanks. Stop after the extended characters have been displayed once.

Code:

```
.model small
.stack 100h
.code
main:
    mov cx,80h        ; start from 80h
    mov bl,0          ; counter

next:
    mov dl,cl
    mov ah,02h
    int 21h

    mov dl,' '        ; space
    int 21h

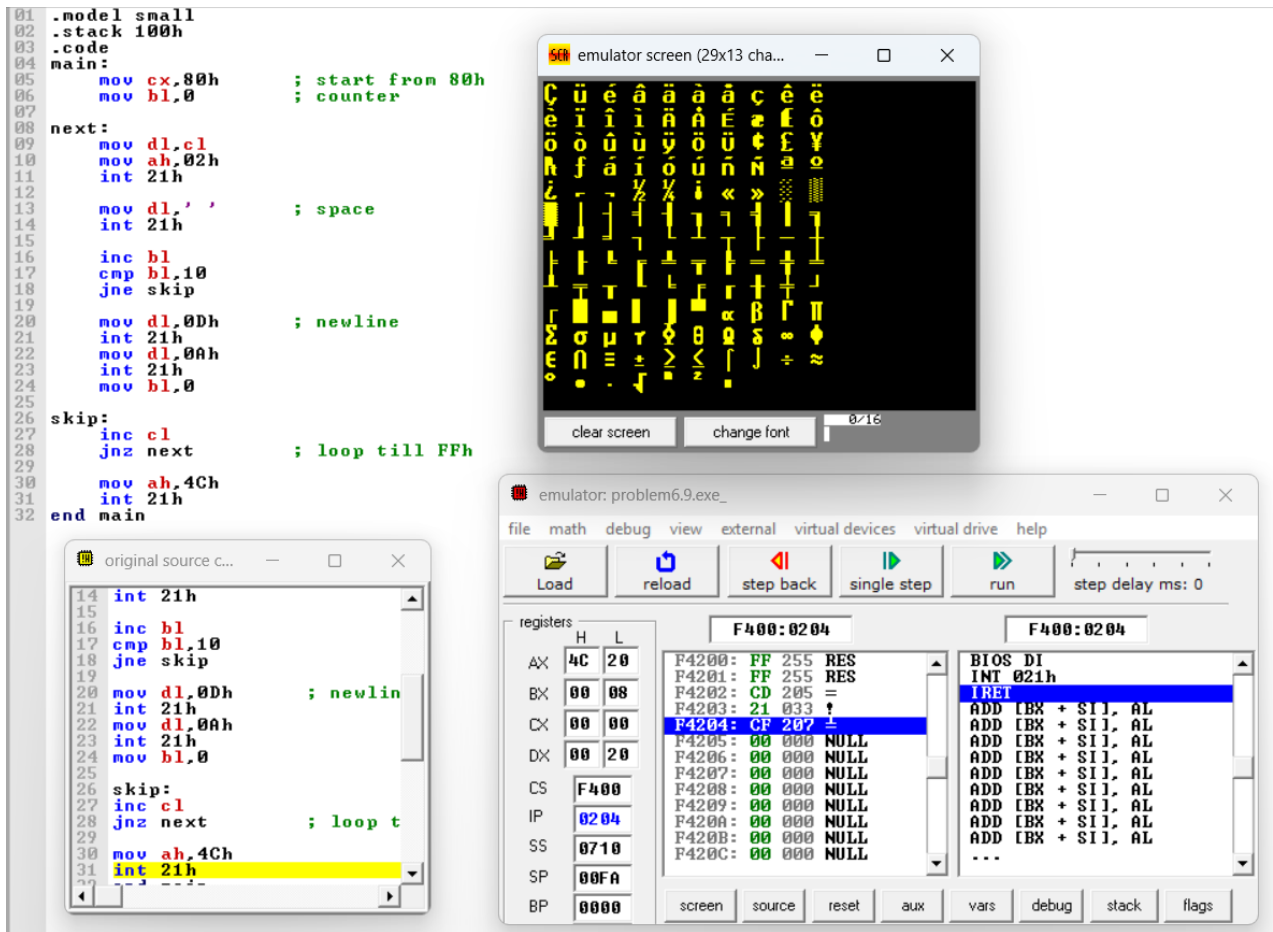
    inc bl
    cmp bl,10
    jne skip

    mov dl,0Dh        ; newline
    int 21h
    mov dl,0Ah
    int 21h
    mov bl,0

skip:
    inc cl
    jnz next          ; loop till FFh

    mov ah,4Ch
    int 21h
end main
```

Output:



Problem No. 6.10

Problem Statement:

Write a program that will prompt the user to enter a hex digit character ("0"..."9" or "A"..."F"), display it on the next line in decimal, and ask the user if he or she wants to do it again.

If the user types "y" or "Y", the program repeats; if the user types anything else, the program terminates. If the user enters an illegal character, prompt the user to try again.

ENTER A HEX DIGIT: 9

IN DECIMAL IS IT 9

DO YOU WANT TO DO IT AGAIN? y

ENTER A HEX DIGIT: C

ILLEGAL CHARACTER - ENTER 0..9 OR A..F: C

IN DECIMAL IT IS 12

DO YOU WANT TO DO IT AGAIN? N

Code:

```
.model small
.stack 100h
.data
msg1 db 13,10,'ENTER A HEX DIGIT: $'
```

```

msg2 db 13,10,'IN DECIMAL IT IS: $'
msg3 db 13,10,'ILLEGAL CHARACTER - ENTER 0..9 OR A..F: $'
msg4 db 13,10,'DO YOU WANT TO DO IT AGAIN? $'

.code
main:
    mov ax, @data
    mov ds, ax
start:
    ; print prompt
    lea dx, msg1
    mov ah, 09h
    int 21h
read_char:
    mov ah, 01h
    int 21h
    mov bl, al

    ; check 0-9
    cmp bl, '0'
    jl invalid
    cmp bl, '9'
    jle valid_num

    ; check A-F
    cmp bl, 'A'
    jl invalid
    cmp bl, 'F'
    jle valid_hex
invalid:
    lea dx, msg3
    mov ah, 09h
    int 21h
    jmp read_char

valid_num:
    sub bl, '0'
    jmp print_dec

valid_hex:
    sub bl, 55          ; 'A' = 65 ? 10 = 65-55

print_dec:
    lea dx, msg2
    mov ah, 09h
    int 21h

    ; convert number in BL to decimal print
    mov al, bl
    cmp al, 10
    jl one_digit

    ; for 10-15
    mov ah, 0

```



```

mov cl, 10
div cl          ; AL=1, AH=remainder
add al, '0'
mov dl, al
mov ah, 02h
int 21h

```

```

mov dl, ah
add dl, '0'
mov ah, 02h
int 21h
jmp ask_again

```

```

one_digit:
add bl, '0'
mov dl, bl
mov ah, 02h
int 21h

```

```

ask_again:
lea dx, msg4
mov ah, 09h
int 21h

```

```

mov ah, 01h
int 21h

```

```

cmp al, 'y'
je start
cmp al, 'Y'
je start

```

```

mov ah, 4Ch
int 21h

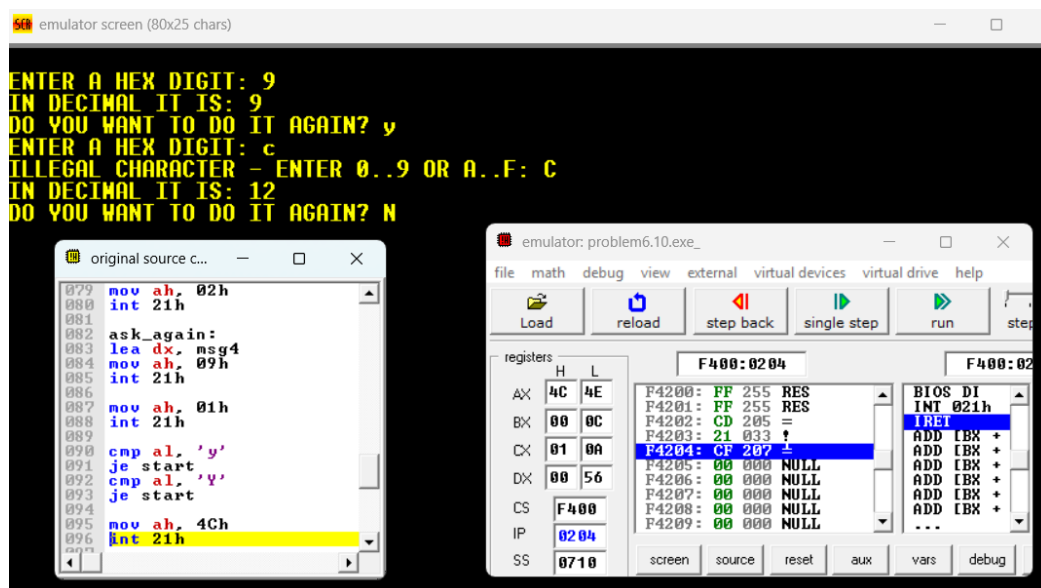
```

```

end main

```

Output:



Problem No. 6.11

Problem Statement:

Do programming exercise 10, except that if the user fails to enter a hex-digit character in three tries, display a message and terminate the program.

Code:

```
.model small
.stack 100h
.data
msg db ' ->Invalid!','$'

.code
main:
    mov ax,@data
    mov ds,ax

    mov cx,3            ; 3 tries

again:
    mov ah,01h          ; input
    int 21h

    ; check '0'-'9'
    cmp al,'0'
    jb checkAF
    cmp al,'9'
    jbe valid

checkAF:
    cmp al,'A'
    jb invalid
    cmp al,'F'
    jbe valid

invalid:
    loop again          ; try again

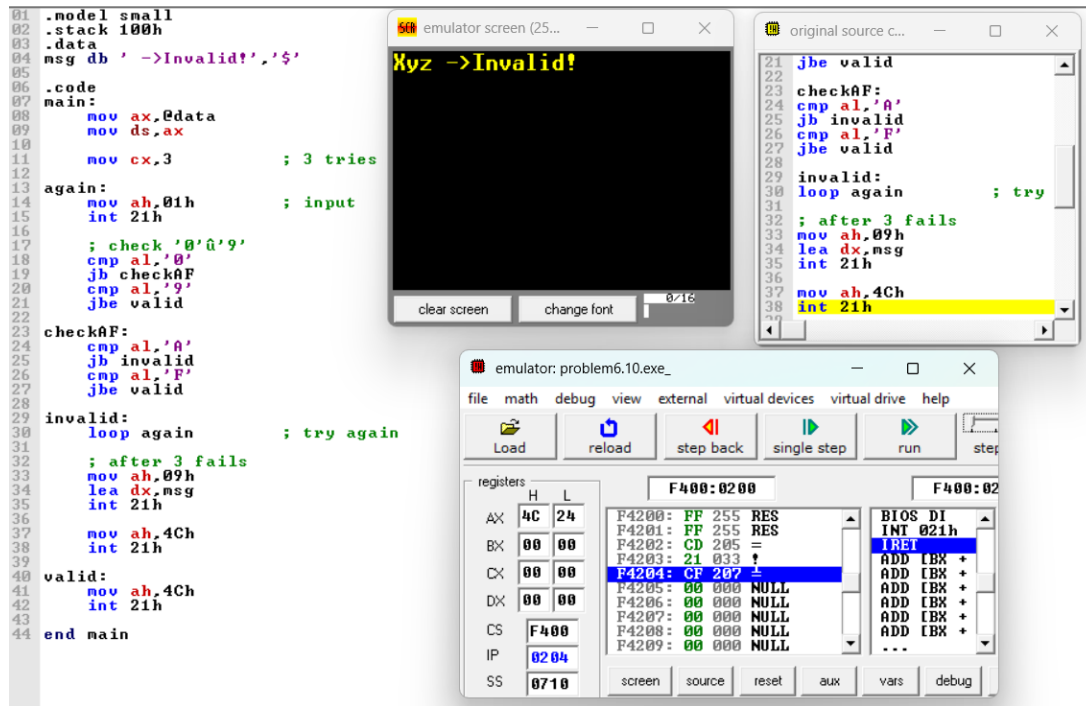
    ; after 3 fails
    mov ah,09h
    lea dx,msg
    int 21h

    mov ah,4Ch
    int 21h

valid:
    mov ah,4Ch
    int 21h

end main
```

Output:



Problem No. 6.12

Problem Statement:

Write a program that reads a string of capital letters, ending with a carriage return, and displays the longest sequence of consecutive alphabetically increasing capital letters read.

Sample execution:

ENTER A STRING OF CAPITAL LETTERS:

FGHADEFGHC

THE LONGEST CONSECUTIVELY INCREASING STRING IS:

DEFGH

Code:

```
.model small
.stack 100h
.data
str db 100 dup('$')
msg db 0Dh,0Ah,'Result: $'

.code
main:
    mov ax,@data
    mov ds,ax
```

```

    ; input
    lea si,str
in1:
    mov ah,1
    int 21h
    cmp al,0Dh
    je done
    mov [si],al
    inc si
    jmp in1

done:
    mov byte ptr [si],'$'

    lea si,str
    mov bx,si        ; best start
    mov di,si        ; current start

    mov cl,1         ; current len
    mov ch,1         ; best len

next:
    mov al,[si]
    cmp al,'$'
    je print

    mov dl,[si+1]
    cmp dl,'$'
    je lastcheck

    mov ah,al
    inc ah
    cmp dl,ah
    je incseq

    ; reset
    inc si
    mov di,si
    mov cl,1
    jmp next

incseq:
    inc cl
    cmp cl,ch
    jbe cont

    mov ch,cl
    mov bx,di

cont:
    inc si
    jmp next

lastcheck:

```

```

cmp cl,ch
jbe print
mov ch,cl
mov bx,di

```

```

print:
mov ah,9
lea dx,msg
int 21h

```

```

mov si,bx
mov cl,ch

```

```

p:
mov dl,[si]
mov ah,2
int 21h
inc si
dec cl
jnz p

```

```

mov ah,4Ch
int 21h

```

```

end main

```

Output:

